



IMDb Data Analytics

A BigQuery Analysis Project on Movie Performance, TV Longevity, and Career Growth.

ANALYST

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INFRASTRUCTURE

Google Cloud Platform (GCP)

0. ENVIRONMENT

Infrastructure & Connectivity

System Configuration

>_ Python Integration

```
PROJECT_ID = "data-engineer-490818"
client = bigquery.Client()

# Connectivity verification
query = "SELECT * FROM imdb.reviews"
df = client.query(query).to_dataframe()
```

✓ Successfully verified connection to BigQuery Public Data.

Schema Architecture

Mapping relationships between TITLE_BASICS, RATINGS, PRINCIPALS, and AKAS using primary keys.

Analysis Scope

Processing 10M+ titles and 58M+ crew member records at cloud scale.



1. Movie Performance Analysis

Top Rated Movies (Votes > 10,000)

Title	Year	Rating
The Shawshank Redemption	1994	9.3
The Godfather	1972	9.2
The Chaos Class	1975	9.2
Attack on Titan: Last Attack	2024	9.2
The Dark Knight	2008	9.1

Genre Efficiency

Documentary / News	9.6 Avg
Romance / War	9.4 Avg
Action / Adventure	9.4 Avg

"Recent high ratings are primarily driven by specialized genres, showing a shift in audience valuation towards factual content."



| 2. Talent & Comedy Dominance



Rajkumar

185 Movies

Highest participation in 8.5+ rated films.

Most Prolific Comedy Actors

Brahmanandam

292 Titles

Jagathy Sreekumar

230 Titles

Innocent Vareed

154 Titles

| 3. TV Series Longevity Analysis

27.96

Average Seasons

Lifespan of top-rated series.

✔ "Masterpieces" sustain decades of demand.

Longest Running Series

Series Title	Seasons
Sesame Street	56 Seasons
Detective Conan	54 Seasons
Survivor	52 Seasons
Saturday Night Live	51 Seasons
The Simpsons	40 Seasons

| 4. Original vs. Sequel Performance

Original Movies (11,491)

6.60 Avg

Sequel Movies (815)

6.32 Avg

★ High Consistency Franchises

Kill Bill

8.33

Demon Slayer

8.12

Avengers

8.03

| 5. Career Trajectory Case Study

Uwe Boll: Rating Improvement



Trend line confirms continuous artistic development.

Leonardo DiCaprio Case

Observed career growth characterized by increasingly selective project filtering and rising ratings.

General Trend

Top directors generally show a **positive rating slope** as experience accumulates.

CONCLUSION

Project Summary & Findings

| Core Analytical Findings



Data-Driven Storytelling

Quantifying cinematic history through petabyte-scale cloud infrastructure on GCP BigQuery.

Survivor Bias

Older classics (6.90) score higher than recent titles (6.44) due to cultural filtering over time.

Sequel Decay

Sequels typically face a rating penalty (~4.2%) compared to the original creative output.

Artistic Mastery

Statistical modeling confirms that top-tier talent generally improves over decades of work.

Questions?

Thank you for your attention.



Huang Xinyi | IMDb BigQuery Analysis Project

| Image Sources



<https://i.ytimg.com/vi/oRs-A3OwRKg/hqdefault.jpg>

Source: www.youtube.com



https://miro.medium.com/O*dIY0cLHzQ6r0dvw0.png

Source: medium.com



<https://en.tourismus.li/assets/images/a/1f53a87126c49b33-t95dhxr5je9v3wm.jpg>

Source: en.tourismus.li



https://media.istockphoto.com/id/176507980/photo/mid-adult-man-directing-film-in-outdoor-movie-set.jpg?s=612x612&w=0&k=20&c=_ma6G3YoKv--UvPt31cTb4H6mHuEu0gG9SWqnvBydAA=

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